MSE 435 Group Project

Efficient utilization of examination rooms is a critical operational challenge in outpatient healthcare clinics. Poor scheduling can lead to patient delays, provider idle time, congestion, and underutilized resources. The project consists of modeling, devising a solution methodology, and implementing an optimization engine for **Examination Room Scheduling in an Outpatient Clinic** in a clinic that operates with limited rooms, multiple appointment types, and shared clinical staff

The community healthcare clinic under study operates from **9:00 a.m. to 5:00 p.m.** and delivers patient services through a mix of clinical appointments and community-based services. Each appointment, characterized by **Appointment type**, **Duration, healthcare provider (Doctor or Nurse Practitioner)** must be assigned to a time slot and an exam room, while accounting for room capabilities, staffing constraints, and operational soft and hard constraints.

Relevant Data:

I. Operating hours:

- Morning admin. time: 9:00-9:30 Monday to Thursday; 8-8:30 on Friday.
- Afternoon admin. time: 4:30pm-5pm Monday to Thursday; 3-3:30 on Friday.
- Noon admin time: 11:30-12:00 every day
- Lunch break-Clinic closed: 12:00-1:00 daily

II. **Examination Rooms:**

Examination Room (ER) Proximity Index (rectilinear distances between Examination Rooms in meters).

ER2	ER3	ER4	ER5	ER6	ER7	ER8	ER9	ER10	ER11	ER12	ER13	ER14	ER15	ER16	
0.5	1.5	2.5	4	4.7	2	3	4.7	4.1	3.3	1.5	3	6.1	11.5	12	**ER1**
	1	2	3.5	4.2	1.5	2.5	4.2	3.6	2.8	2	3.5	5.6	11	11.5	**ER2**
		1	2.5	3.2	0.5	1.5	3.2	2.6	1.8	3	4.5	4.6	10	10.5	**ER3**
			1.5	2.2	1.5	1.5	2.2	1.6	2.8	4	5.5	3.6	9	9.5	**ER4**
				0.7	3	3	2.5	3.1	4.3	5.5	6	3.9	7.5	8	**ER5**
					3.7	3.7	3.2	3.8	5	6.2	6.7	4.6	6.8	7.3	**ER6**
						1	2.7	2.1	1.3	2.5	4	4.1	9.5	10	**ER7**
							1.7	1.1	1.3	2.5	4	3.1	8.5	9	**ER8**
								0.6	1.8	3.2	5.7	1.4	6.8	7.3	**ER9**
									1.2	2.6	5.1	2	7.4	7.9	**ER10**
										1.8	4.3	2.8	8.2	8.7	**ER11**
											2.5	4.6	10	10.5	**ER12**
												7.1	12.5	13	**ER13**
													5.4	5.9	**ER14**
														0.5	**ER15**

III. Healthcare Providers

- Available certain days of the week- certain portions of the day. (Check the data)

IV. Appointments (Check the data)

Characterized by:

- Appointment id
- Patient id
- Duration
- Healthcare provider

V. Other factors

- No-shows.
- Uncertainty in appointment times.

Your **task** is to

1. Understand the problem and provide mathematical model(s). Try to use a mix of formulations that will handle large sizes (hint: column generation). Clearly define the decision variables, the objective function, and the constraints.
2. Devise a solution methodology. Use decomposition concepts learned in class and/or heuristic methods to solve the problem.
3. Solve the resulting model. If necessary, refine the approach to account for practical constraints and to provide meaningful schedules.
4. Validate the approach, e.g. by comparing different policies
    a. Assign a Healthcare Provider to a single room for the entire day or week
    b. Assign a HP to a cluster of rooms (within a certain proximity) .
    c. Block certain days for certain healthcare providers
    d. Use certain admin. time for extenuating circumstances
    e. Overbook, taking no shows into account.
    f. Account for uncertainty in examination durations.
5. Carry out testing and comparison. Provide different schedules. Come up with KPI measure and compare. Provide tables/graphs comparing different approaches and schedules.
6. Write a final report.

Deliverables and deadlines

You are required to provide a schedule (in a visual format, for example as a Gantt chart).

A good solution is one that

- Assigns all appointments and respects all constraints.
- Has a compact schedule for Healthcare Providers.
- Minimizes swaps between exam rooms (travel distance preferably).
- Accounts for uncertainty and no-shows.

To help you, the deliverables are divided among two progress reports, one final report, and a presentation

- Progress Report 1: Task components **1 to 2**. Due Friday, **Feb. 27th, 2026@ 11:59 pm ET.**
- Progress Report 2: Task components **1 to 4**. Due Friday**, March 20th, 2026@ 11:59 pm ET.**
- Final Report: Task components **1 to 6.** Due Friday, April 3rd, 2026@ 11:59 pm ET. Crowdmark will stay open till April 10th, 2026.
- In-class presentation on **Wednesday, April 1st, 2026**. The order of presentations will be revealed then. A copy of the presentation is due by midnight on **March 31st (strict deadline)** via dropbox.

Grading:

The **grading** of the project will depend on

- the sophistication and quality of the proposed approach(es).
- the extent to which you applied material learned in the course.
- the quality of the implementation and testing.
- the quality of the presentation and written reports.
- Report 1 is worth
- Report 2 is worth

Grading scheme:

- Progress Report 1: 8%
- Progress Report 2: 8%
- Presentation: 9%
- Report: 8%

Progress Reports 1 and 2 are worth 8% each, but their grade is finalized in the final report, i.e. you are expected to keep improving them till the final report.

Important Notes:

- References to external material, if any, should be included.
- Be concise and to the point. Detailed output, code listings, etc. should be put in an appendix. Only provide summary results in the body of the report (e.g. in table or figure formats).
- Use 12 pt font with at least 1-inch margins all around.
- Submit your codes with the final report.

You are bound by policy 71 and the course policy on the use of Gen. AI. Violations will not be tolerated.